$$I = \int_{-10}^{10} f(x)\,dx = 10\int_{-1}^{1} f(10t)\,dt$$

$$I_{app} = 10\sum_{i=1}^{n} w_i f_e(10t_i)$$

$f_e(x) = 3{,}5x^2\cos(2x)$

$I_{exact} \approx 332{,}22$

n=4:

$t_i: -1, -\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 1$

$w_i: \frac{1}{6}, \frac{5}{6}, \frac{5}{6}, \frac{1}{6}$

$f_e(10) = 3{,}5 \cdot 100 \cdot \cos(20) \approx 142{,}835$

$f_e\left(\frac{10}{\sqrt{5}}\right) = 3{,}5 \cdot \frac{100}{5} \cdot \cos\left(\frac{20}{\sqrt{5}}\right) \approx -63{,}448$

$I_4 = 10\left(2 \cdot \frac{1}{6} \cdot 142{,}835 + 2 \cdot \frac{5}{6} \cdot (-63{,}448)\right) \approx -58{,}135$

n=5:

$t_i: -1, -\sqrt{\frac{3}{7}}, 0, \sqrt{\frac{3}{7}}, 1$

$w_i: \frac{1}{10}, \frac{49}{90}, \frac{32}{45}, \frac{49}{90}, \frac{1}{10}$

$f_e(0) = 3{,}5 \cdot 0 \cdot \cos(0) = 0$

$f_e(10) = 142{,}835$

$f_e\left(10\sqrt{\frac{3}{7}}\right) = 3{,}5 \cdot 100 \cdot \frac{3}{7} \cdot \cos\left(20\sqrt{\frac{3}{7}}\right) = 142{,}08$

$I_5 = 10\left(\frac{32}{45} \cdot 0 + 2 \cdot \frac{1}{10} \cdot 142{,}835 + 2\,\frac{49}{90} \cdot 142{,}08\right)$

$I_5 \approx 1832{,}76$

Оценка:

n=4: $\Delta I_4 \approx 913{,}57$

n=5: $\Delta I_5 \approx 1500{,}54$